(4) 64 TiB - VM => $2^6 \times 2^{40} = 2^{46}$ B

a) 8 GiB - PM => $2^3 \times 2^{30} = 2^{33}$ B

byte addressing

16 KiB pages => $2^4 \cdot 2^{10} = 2^{14}$ B

32-entry TLB (2-way SA)

256 KiB Data Cache (2-way SA) = 2^{18} B = $\frac{2^{18}}{2^7} = \underbrace{2^{11} \text{ blocks}}$

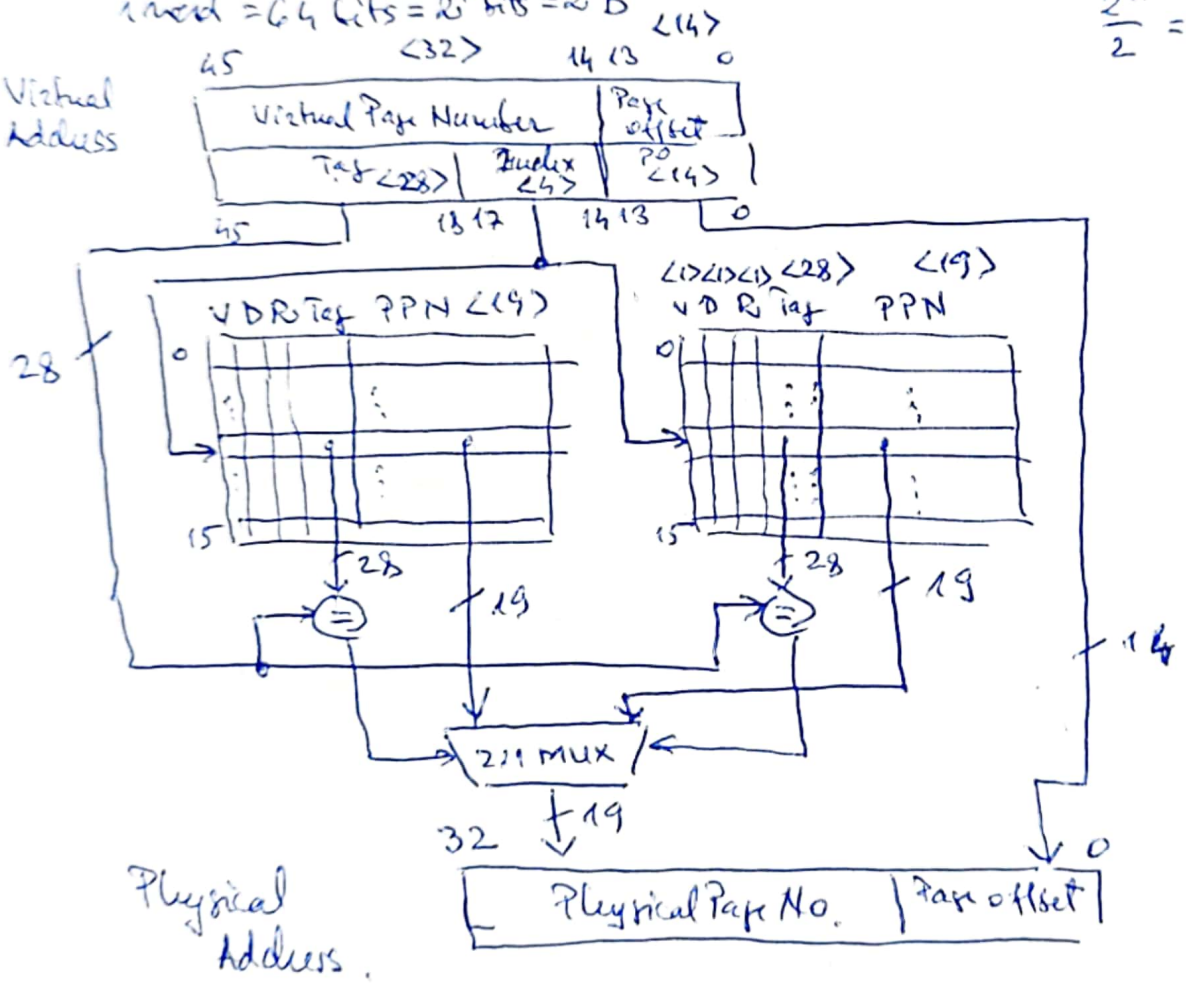
$\Downarrow$

$\frac{2^{11}}{2} = 2^{10}$ blocks / bank

1 block = 16 words = 2^4 W = $2^4 \cdot 2^3$ B = 2^7 B

1 word = 64 bits = 2^6 bits = 2^3 B

$$\begin{array}{r} 32 - \\ 17 \\ \hline 15 \end{array}$$



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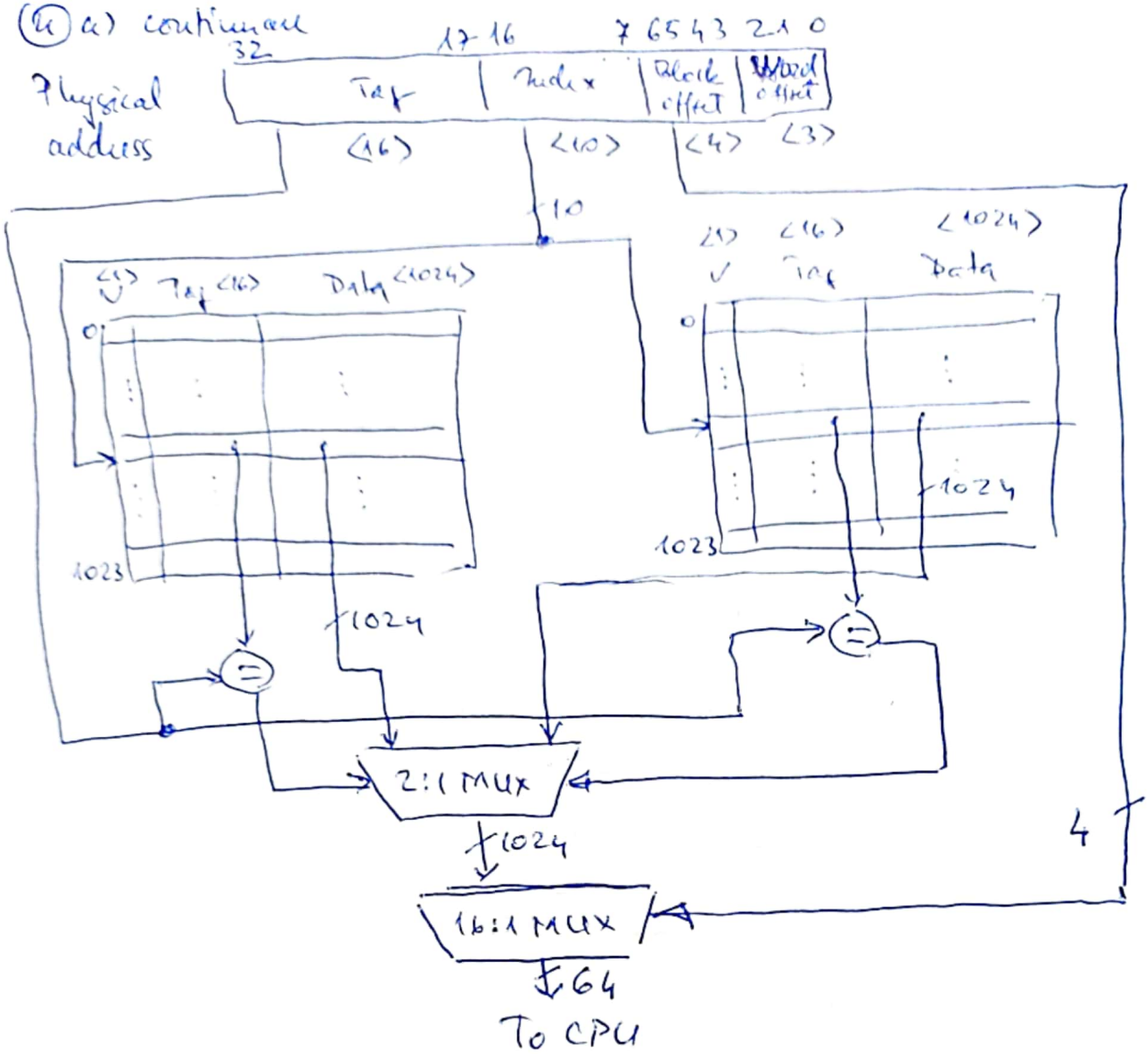
(4) a) continuare



b) TLB size total = $\underbrace{2^5}_{\text{entries}} \times (\underbrace{31 \text{ bits}}_{\text{Tag, V, D, R}} + \underbrace{19 \text{ bits}}_{\text{PPN}}) = 2^5 \times 50 \text{ bits}$

$\cong 2^5 \times 2^6 \text{ bits}$

$= 2^5 \times 2^3 B = 2^8 B$

$= \boxed{0.25 \text{ KiB}}$

Cache size total = $\underbrace{2}_{\text{s.a}} \times \underbrace{2^{10}}_{\text{index}} \times (\underbrace{17 \text{ bits}}_{\text{ctrl bits}} + 2^7 B)$

$= 2^{11} \times 17 \text{ bits} + 2^{18} B$

$\cong \underbrace{2^{11} \times 2^5 \text{ bits}}_{2^{16} \text{ bits} = 2^{13} B = 8 \text{ KiB}} + \underbrace{256 \text{ KiB}}_{\text{data}} \cong \boxed{264 \text{ KiB}}$

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